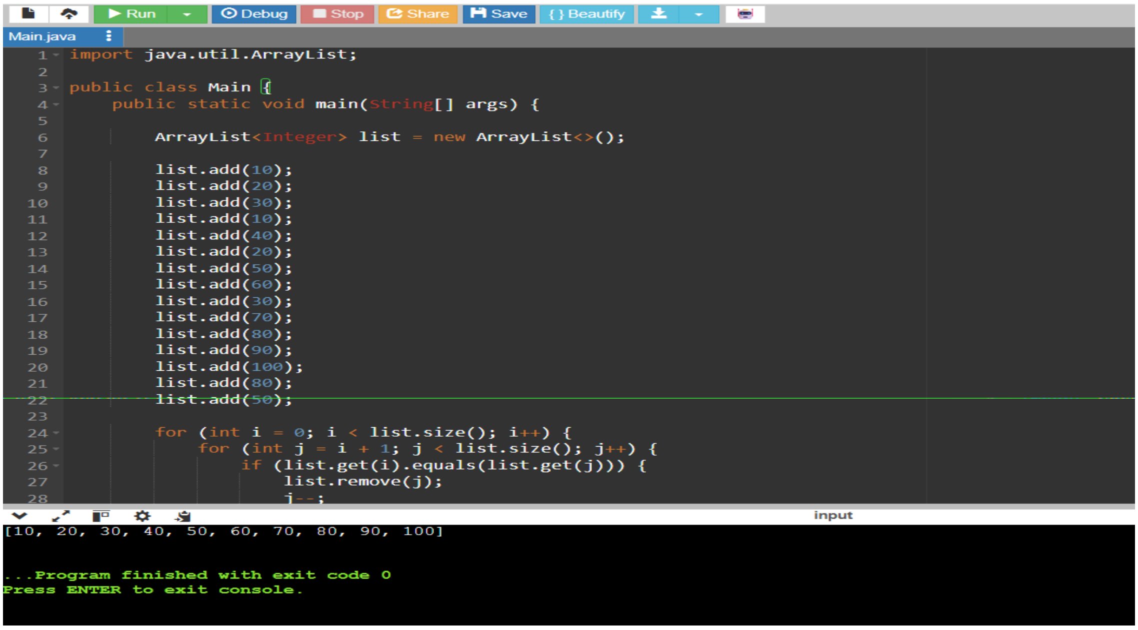
1. Write a program to store 15 integer values in an ArrayList and remove all duplicate elements without using another collection.

```
import java.util.ArrayList;

public class Main {
  public static void main(String[] args) {

    ArrayList<Integer> list = new ArrayList<>();

    list.add(10);
    list.add(20);
    list.add(30);
    list.add(10);
    list.add(40);
    list.add(20);
    list.add(50);
    list.add(60);
    list.add(30);
    list.add(70);
    list.add(80);
    list.add(90);
    list.add(100);
    list.add(80);
    list.add(50);

    for (int i = 0; i < list.size(); i++) {
      for (int j = i + 1; j < list.size(); j++) {
        if (list.get(i).equals(list.get(j))) {
          list.remove(j);
```

```
                j--;
            }
        }
    }

    System.out.println(list);
  }
}
```

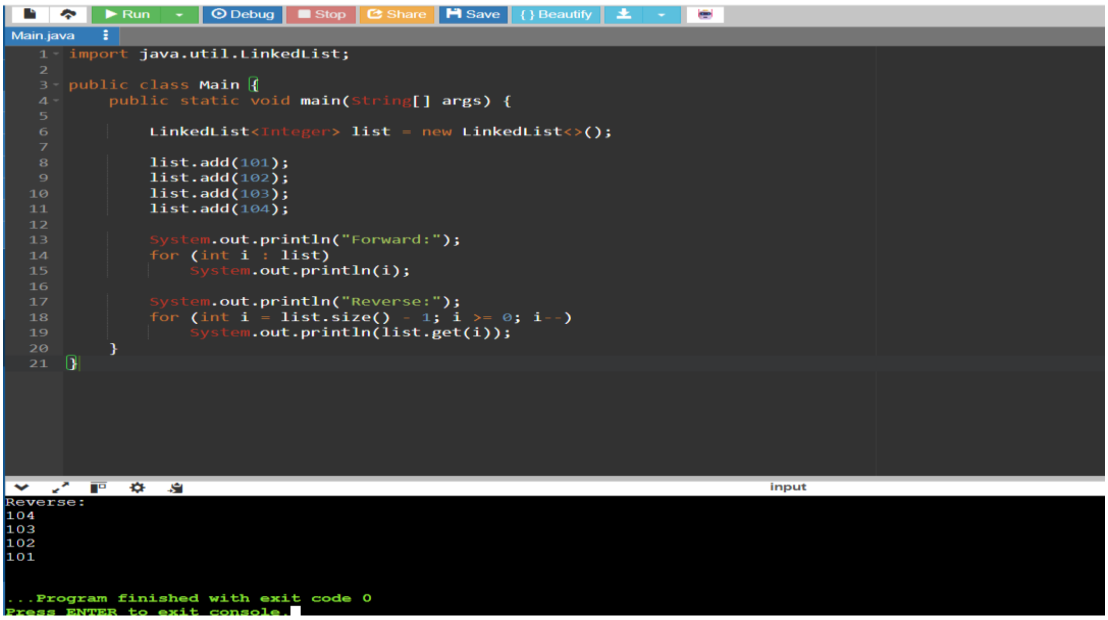
Output



2. Store employee IDs in a LinkedList and display them in both forward and reverse order.

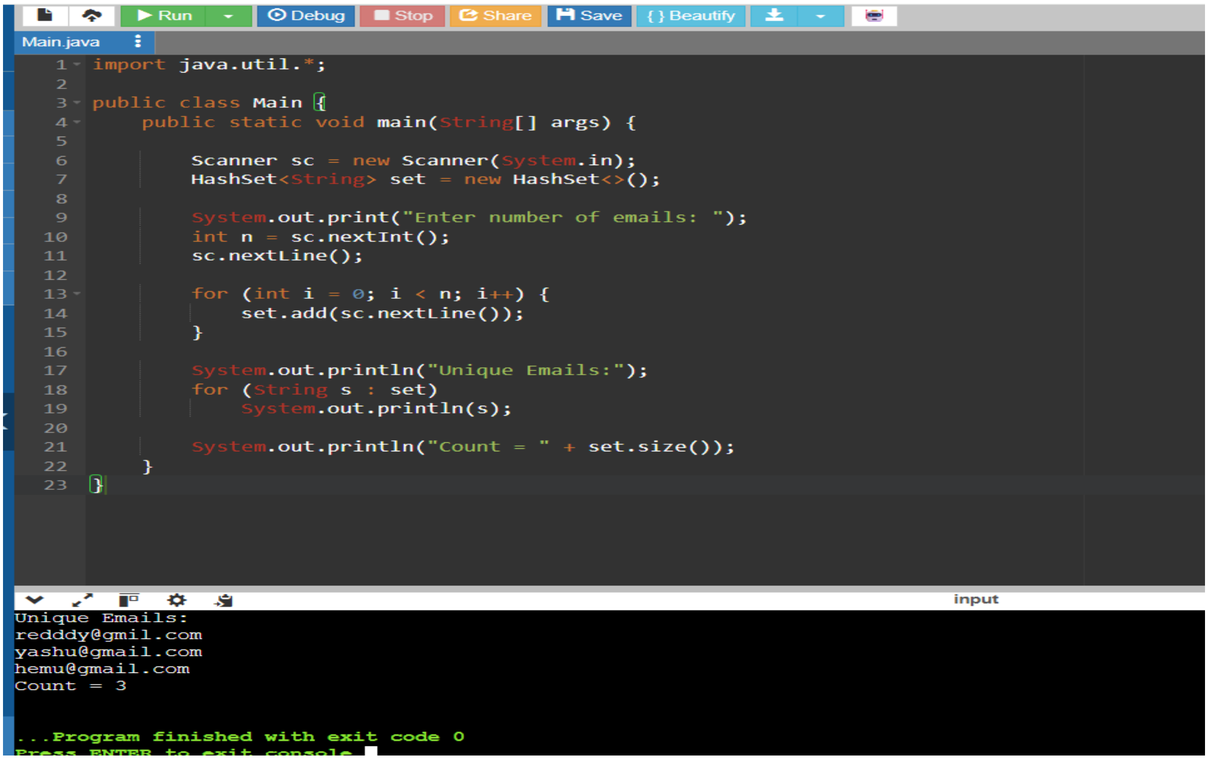
```
import java.util.LinkedList;


public class Main {
    public static void main(String[] args) {


        LinkedList<Integer> list = new LinkedList<>();
```

```
        list.add(101);
        list.add(102);
        list.add(103);
        list.add(104);

        System.out.println("Forward:");
        for (int i : list)
            System.out.println(i);

        System.out.println("Reverse:");
        for (int i = list.size() - 1; i >= 0; i--)
            System.out.println(list.get(i));
    }
}
```



3. Store email IDs entered by the user in a HashSet.Display only unique email IDs and count the total unique entries.

```java
import java.util.*;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        HashSet<String> set = new HashSet<>();

        System.out.print("Enter number of emails: ");
        int n = sc.nextInt();
        sc.nextLine();

        for (int i = 0; i < n; i++) {
            set.add(sc.nextLine());
        }

        System.out.println("Unique Emails:");
        for (String s : set)
            System.out.println(s);

        System.out.println("Count = " + set.size());
    }
}
```



4. Write a Java program to count the frequency of each word in a paragraph using a HashMap.

```
import java.util.*;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter paragraph:");
        String str = sc.nextLine();

        String[] words = str.split(" ");

        HashMap<String, Integer> map = new HashMap<>();

        for (String w : words) {
            if (map.containsKey(w))
                map.put(w, map.get(w) + 1);
            else
                map.put(w, 1);
        }
```

```
        for (String key : map.keySet()) {
            System.out.println(key + " = " + map.get(key));
        }
    }
}
```